

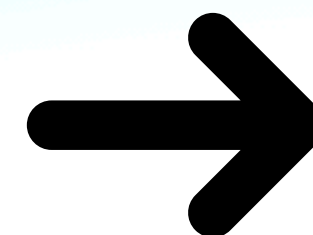
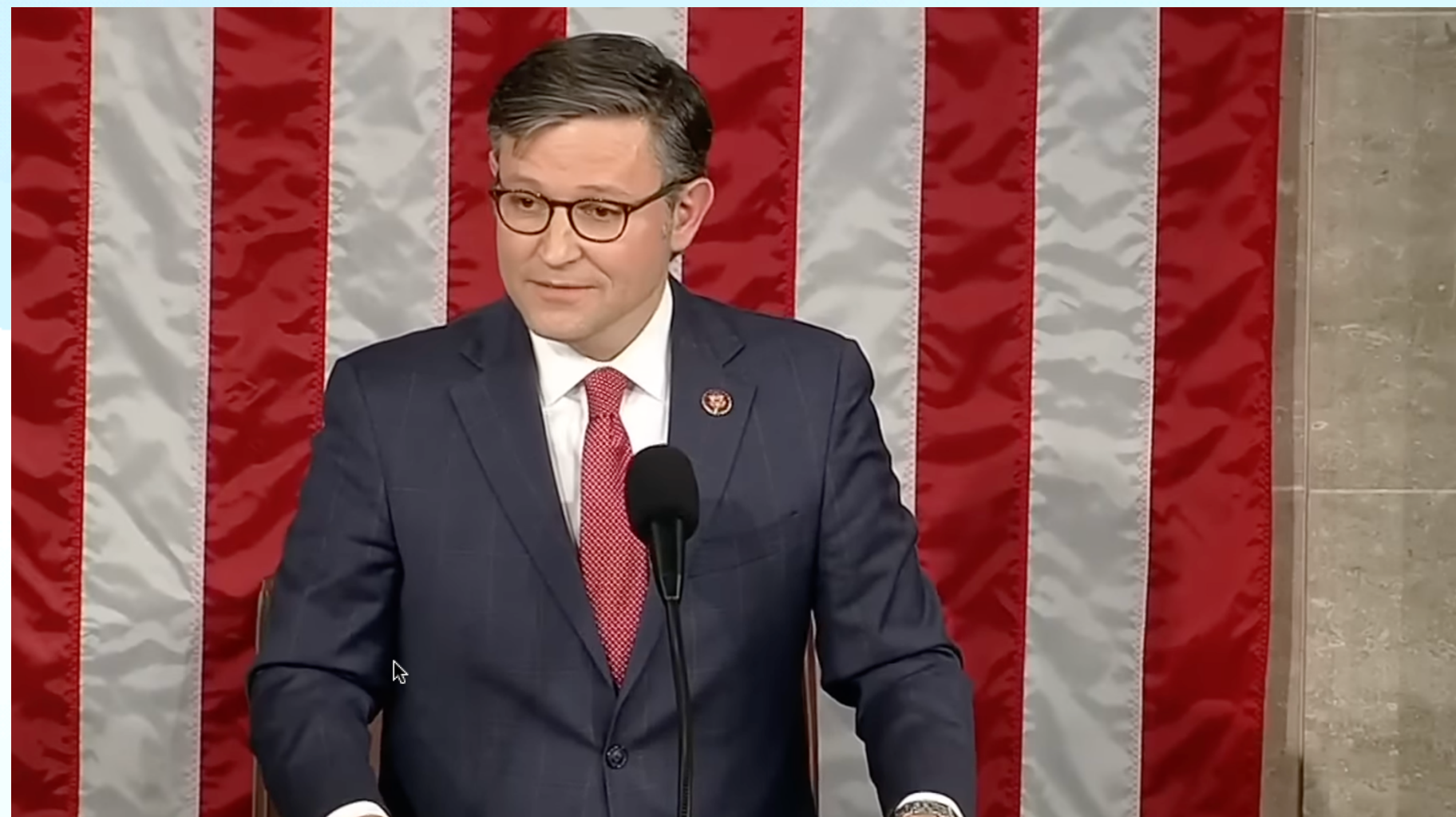
This Floor Speech Will Be Televised

**Understanding the Factors that Influence When Floor Speeches
Appear on Cable Television - Bryce J. Dietrich, Dan Schultz, Tracey Jaquith**

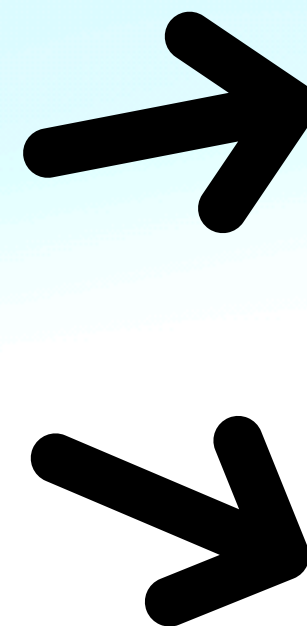
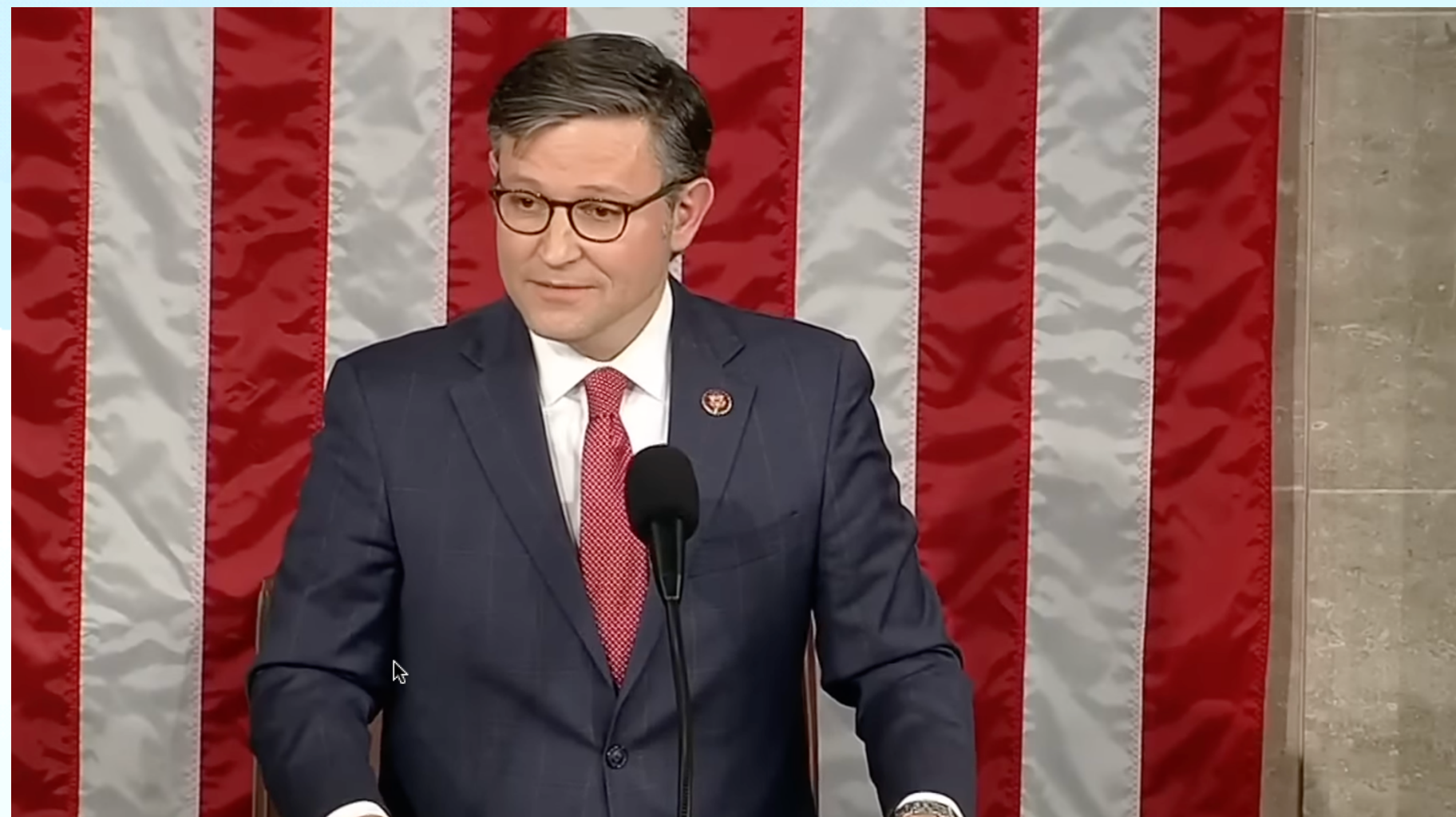
Marek Landert, Joel Asper - 17.11.2023

How do emotional expressions in congressional floor speeches affect the likelihood of being aired on major cable news networks?

Introduction



Introduction



Hypotheses

H1: Cable news organizations are more likely to broadcast floor speeches delivered at **higher vocal pitch**.

H2: Cable news organizations are more likely to broadcast floor speeches which express a negative sentiment.

H3: When members of Congress express negativity towards Congress, their floor speeches are more likely to be aired by cable news organizations.

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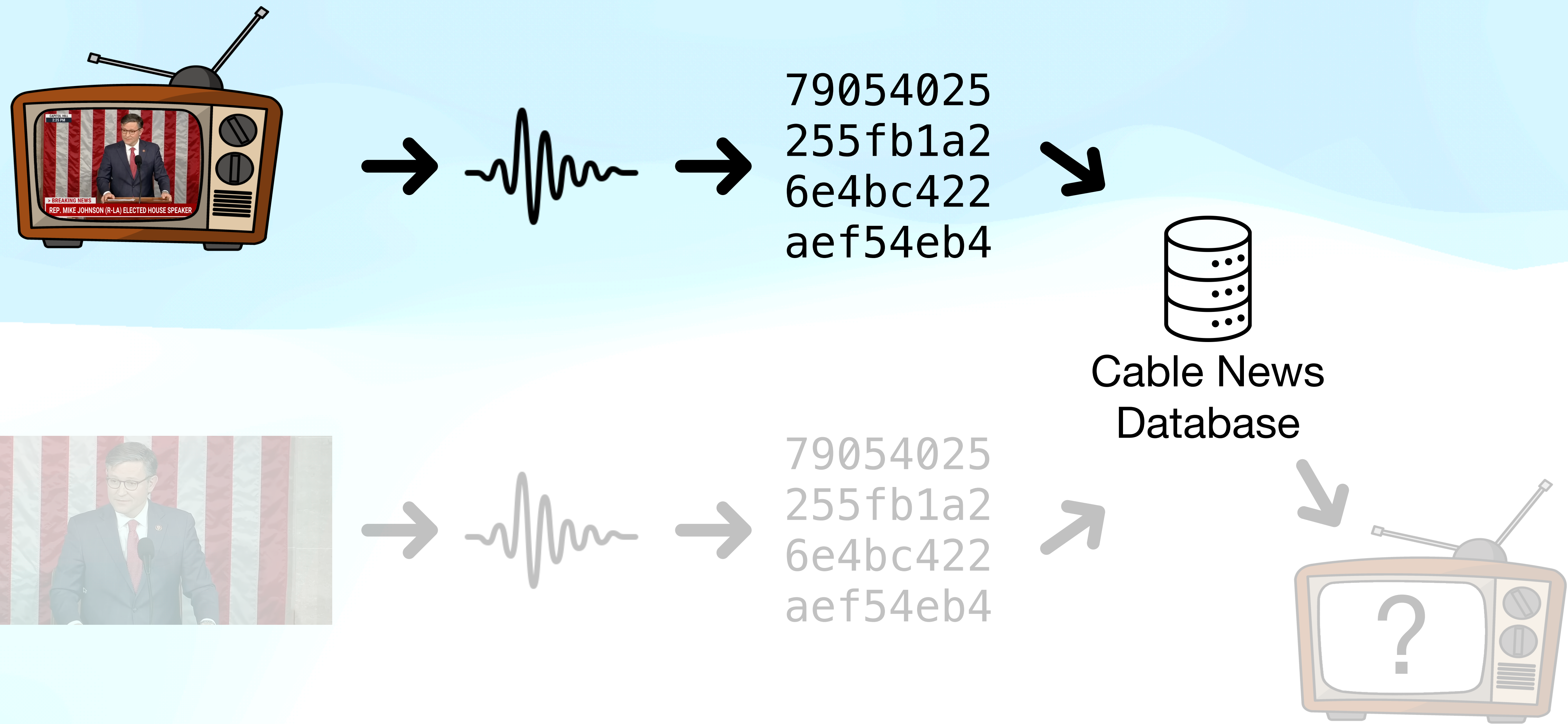
Hypotheses

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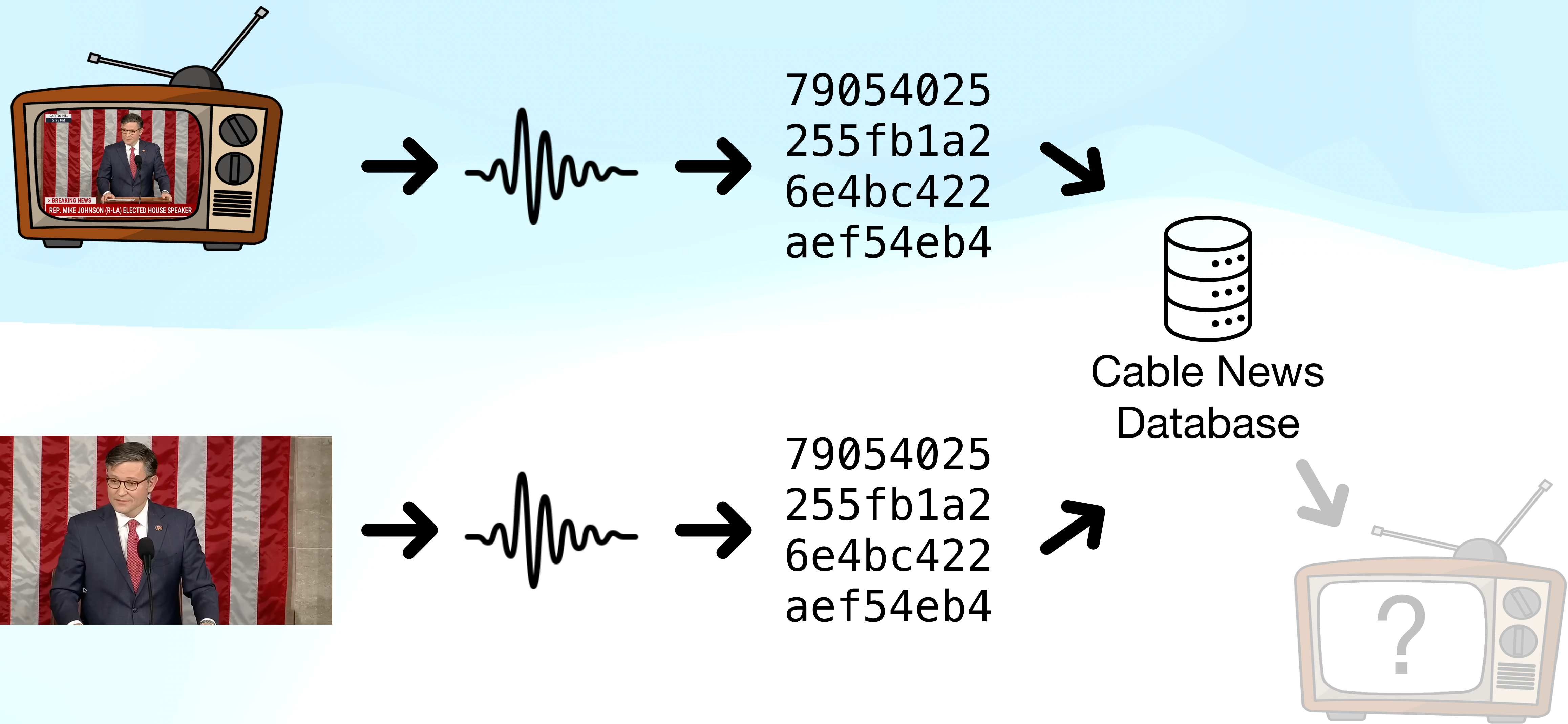
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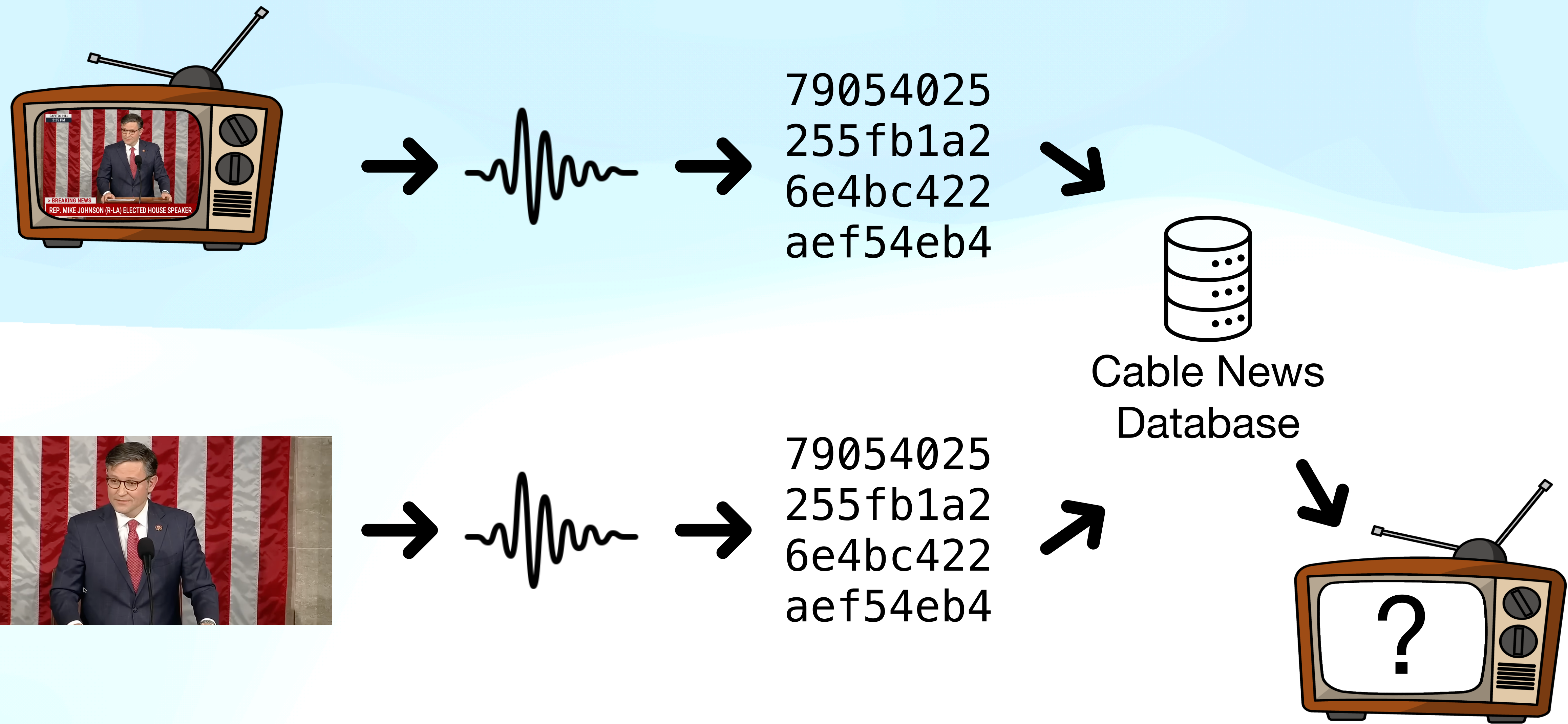
Methodology



Methodology



Methodology



Methodology - Data Collection

- **Audio Data:** 6,432 hours of audio from 74,158 floor speeches from HouseLive (January 6, 2009 to August 4, 2014)
- **Television Data:** 145,706 hours of cable news broadcasts from CNN, Fox News, and MSNBC

Methodology - Emotional Intensity Measurement

- Vocal pitch as a proxy for emotional intensity in speeches
- Praat to extract vocal pitch
- Standardized vocal pitch against each Member of Congress' average to account for gender and individual differences

Methodology - Sentiment Analysis

- How negative or positive do they speak?
- Google Cloud Natural Language API for sentiment analysis, focusing on overall sentiment
- ...and specific sentiment towards Congress.

Methodology - Control Variables

Dummy variable indicating...

- ...if the speaker is the **speaker of the House, Majority/Minority leader**.
- ...if the speaker is the **chair of a committee**.
- ...if the speech is **less than one minute**.
- ...if the speech is delivered during an **election year**.
- ...whether the speaker is **male or female**.
- ...the **duration** of the speech.
- ...if a speaker covers an issue owned by their **respective party**.

Results

Measure television coverage in four ways:

1. whether a floor speech **appeared on television.**
2. **number of times** the floor speech was televised.
3. **total air time** dedicated to the speech.
4. estimated number of **total viewers.**

Results

Table 1: CNN, Fox News, and MSNBC Coverage of Verbal and Non-Verbal Emotional Expressions on the Floor of the U.S. House of Representatives

	Televised		Total Coverage		Total Minutes		Total Viewers	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Constant	−5.883** (0.074)	−6.013** (0.307)	−5.421** (0.139)	−6.293** (0.433)	−3.512** (0.442)	−3.548** (0.436)	−16.053** (1.834)	−15.724** (2.073)
Vocal Pitch	0.617** (0.057)	0.605** (0.058)	0.857** (0.093)	0.882** (0.101)	0.289** (0.040)	0.281** (0.037)	1.315** (0.194)	1.272** (0.196)
Overall Sentiment	0.237** (0.061)	0.226** (0.058)	0.269** (0.090)	0.318** (0.106)	0.095** (0.033)	0.095** (0.033)	0.497** (0.145)	0.502** (0.141)
Congress Sentiment	0.181† (0.102)	0.177† (0.102)	0.186† (0.106)	0.142 (0.115)	0.076† (0.042)	0.075† (0.045)	0.314† (0.170)	0.310† (0.181)
Republican		−0.332† (0.200)		−0.677† (0.358)		−0.132 (0.112)		−0.853 (0.568)
DW-Nominate		1.197** (0.415)		1.745** (0.604)		0.443* (0.206)		2.242* (1.023)
Party Issue		−0.073 (0.063)		−0.134 (0.101)		−0.030 (0.028)		−0.130 (0.130)
Seniority		0.022** (0.006)		0.016† (0.008)		0.010** (0.004)		0.034* (0.016)
House Leader		1.899** (0.195)		2.090** (0.425)		0.857** (0.202)		3.964** (0.807)
Committee Chair		0.197 (0.223)		0.252 (0.304)		0.099 (0.117)		0.352 (0.562)
Male		−0.222 (0.181)		0.204 (0.297)		−0.075 (0.121)		−0.101 (0.493)
White		−0.195 (0.230)		0.173 (0.382)		−0.065 (0.130)		−0.366 (0.548)
CQ Bills		0.212** (0.051)		0.195† (0.100)		0.109** (0.031)		0.329* (0.138)
One Minute		−0.053 (0.172)		0.070 (0.292)		0.025 (0.063)		−0.254 (0.305)
Duration		−0.116* (0.048)		−0.160* (0.080)		−0.028 (0.019)		−0.274* (0.114)
Election Year		−1.005** (0.174)		−1.015** (0.330)		−0.402** (0.086)		−2.122** (0.373)
N	71,198	71,197	71,198	71,197	71,198	71,197	71,198	71,197
Log-Lik	−1,624.930	−1,541.353	−1,965.667	−1,909.634	−1,627.715	−1,545.823	−1,830.763	−1,750.996
AIC	3,257.860	3,114.706	3,939.335	3,851.267	3265.430	3125.646	3671.525	3535.993

Results

- Vocal pitch 1 std. dev. above baseline
- Floor speeches at higher emotional intensity are more likely to be aired

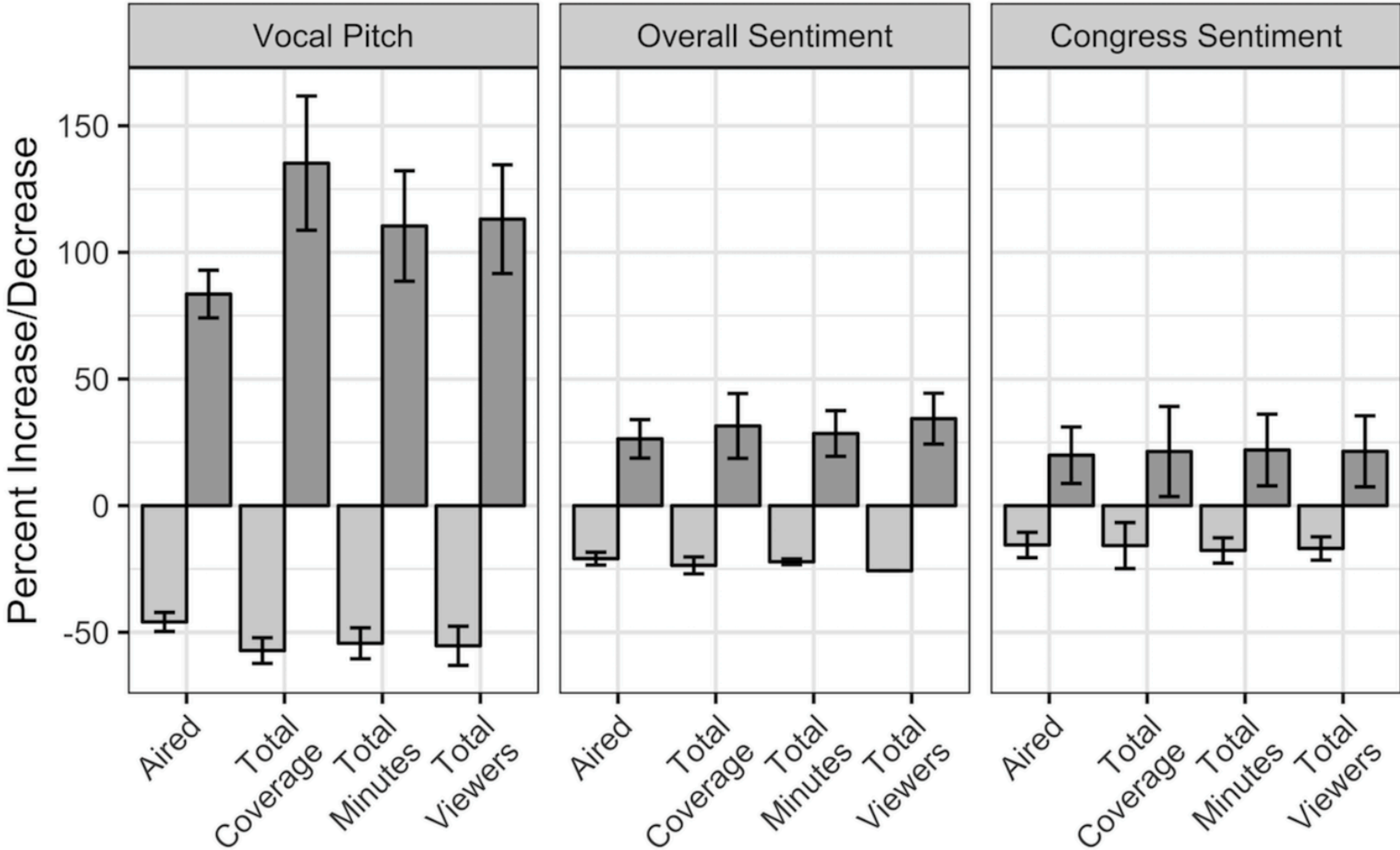
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Results

Hypothesis 1

Hypothesis 2

Hypothesis 3

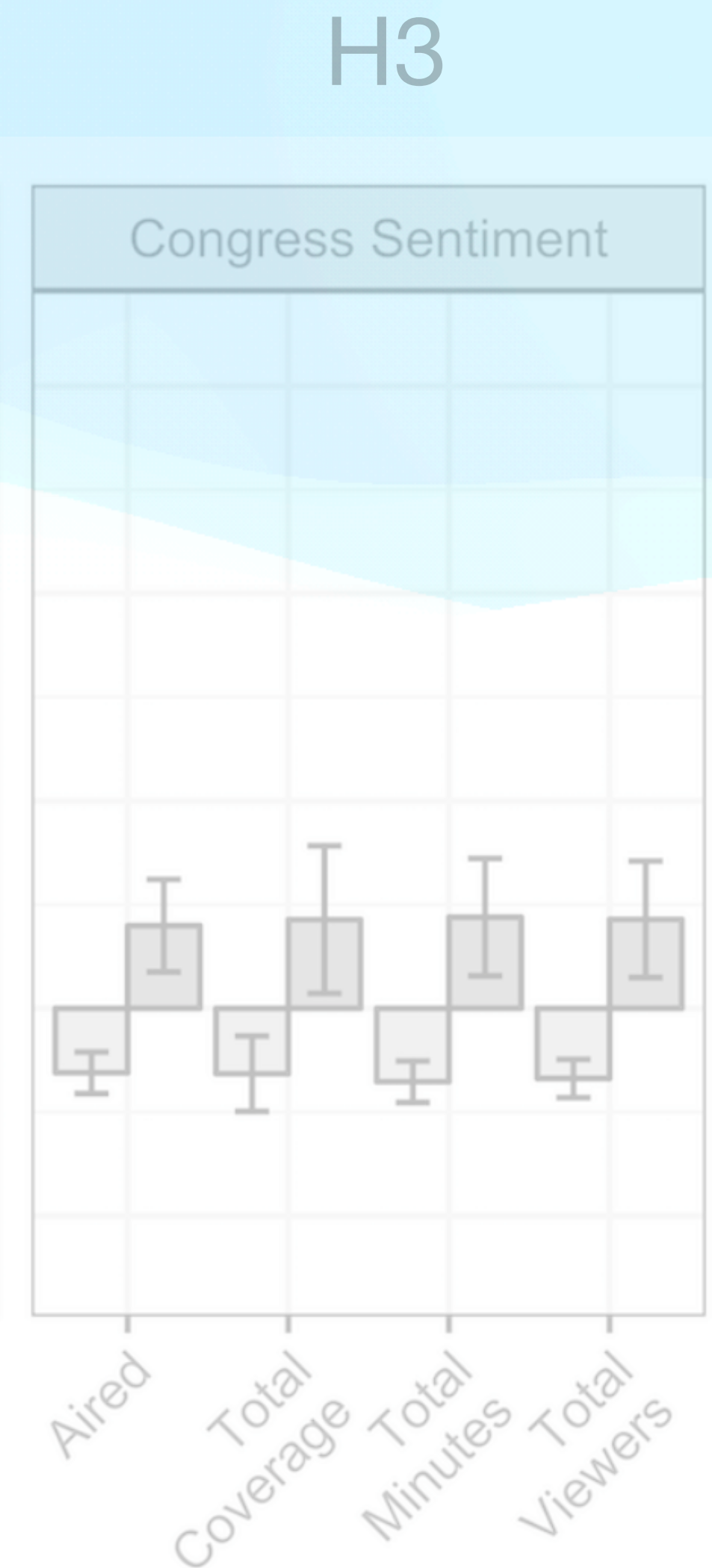
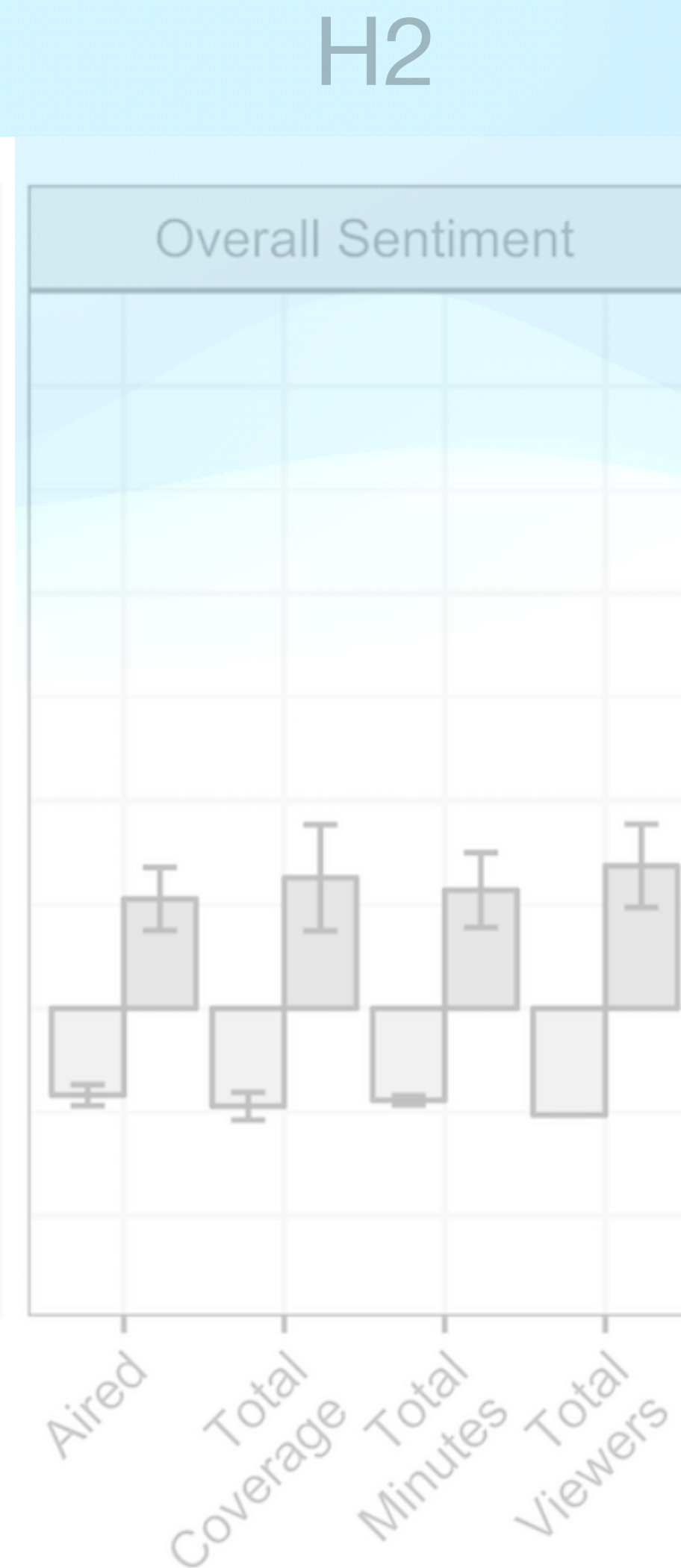
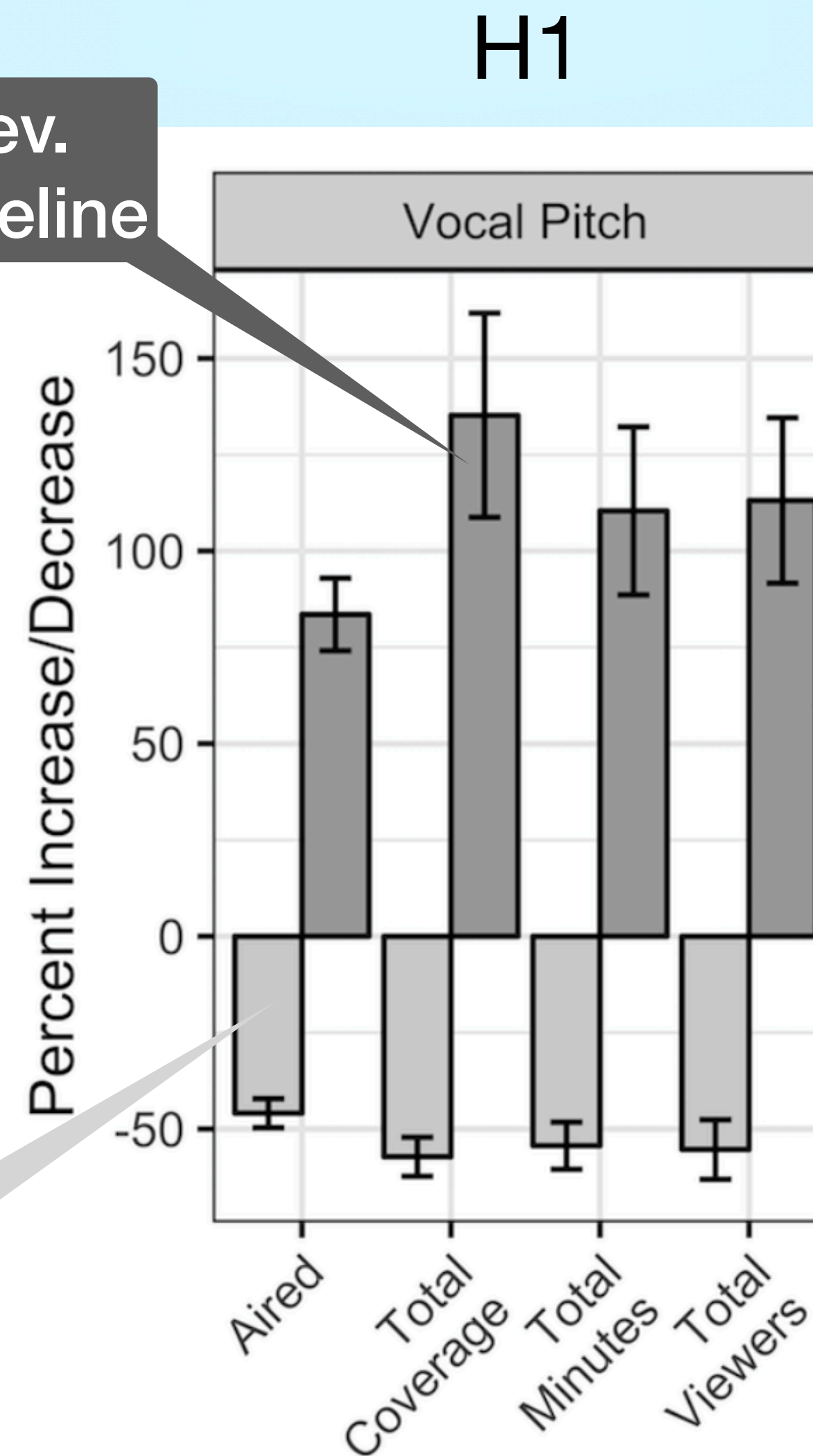


Results - Vocal Pitch

- Emotionally intense speeches are more likely to appear in cable news

1 std. dev.
above baseline

1 std. dev.
below baseline

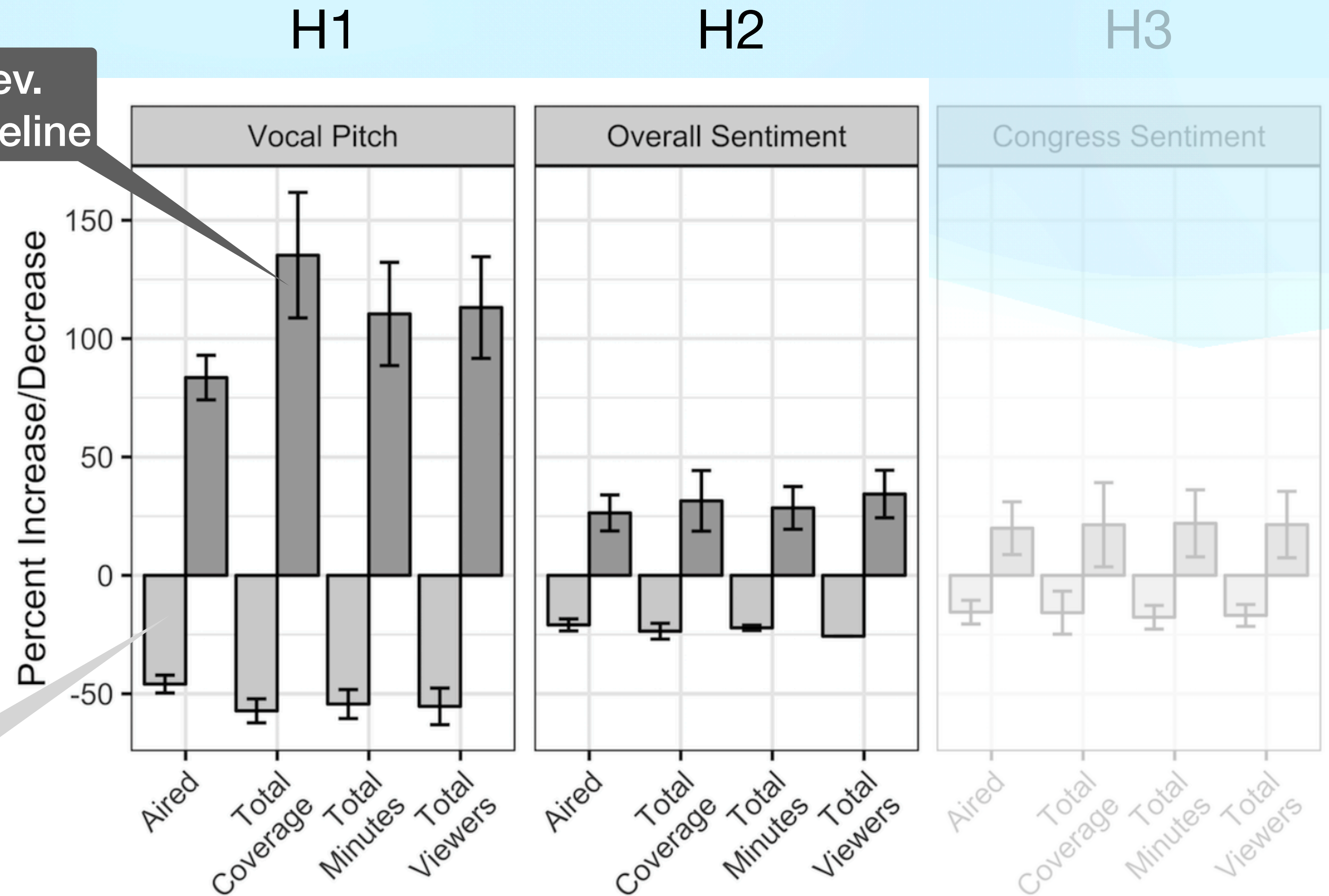


Results - Overall Sentiment

- Speeches with more negative sentiment are more likely to be aired
- Smaller influence than emotional intensity

1 std. dev.
above baseline

1 std. dev.
below baseline

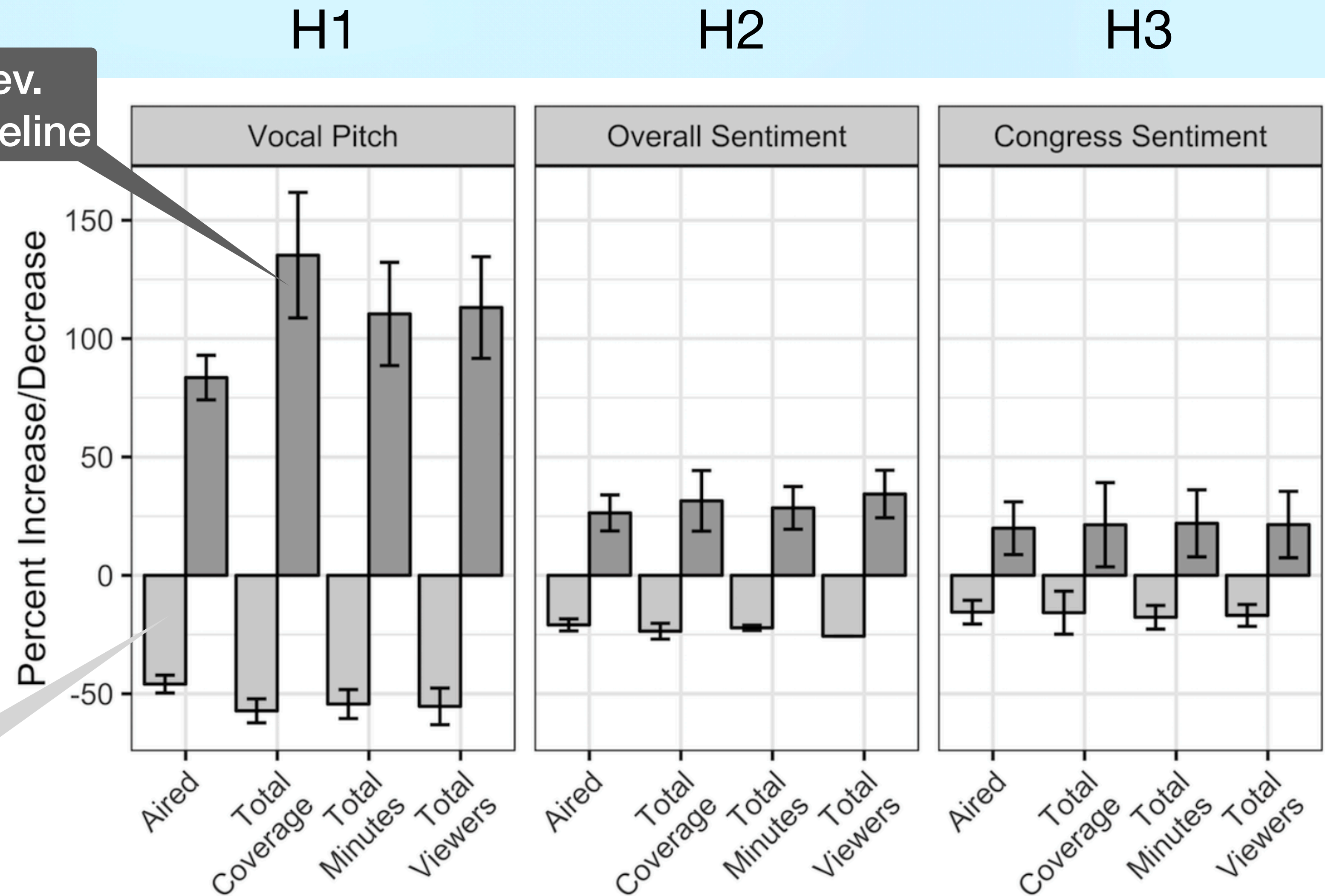


Results - Congress Sentiment

- Similar results for negativity towards Congress as for general negativity
- But results are less statistically significant

1 std. dev.
above baseline

1 std. dev.
below baseline



Discussion - Hypotheses Revisited...

H1: Cable news organizations are more likely to broadcast floor speeches delivered at **higher vocal pitch**.



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Discussion

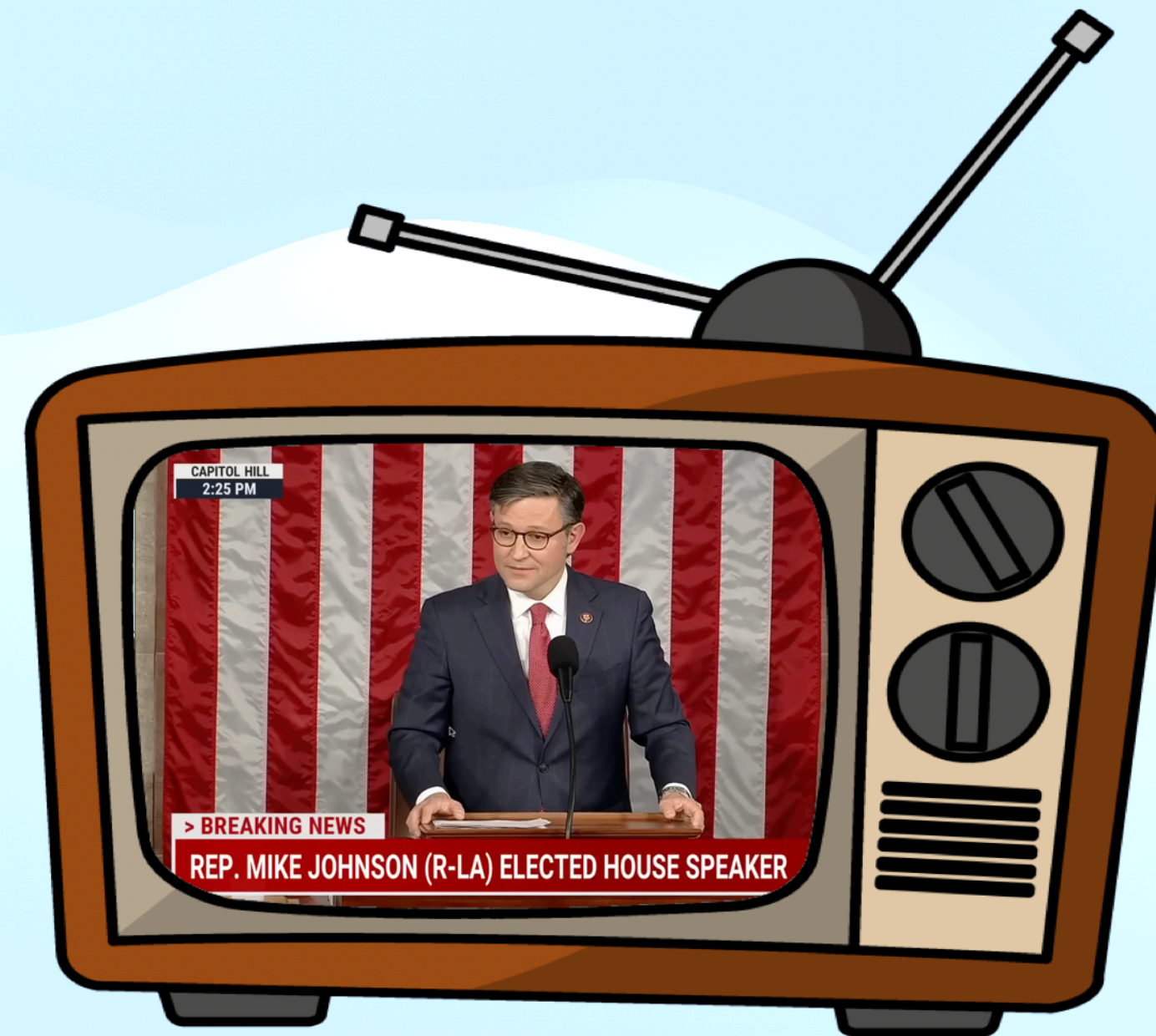
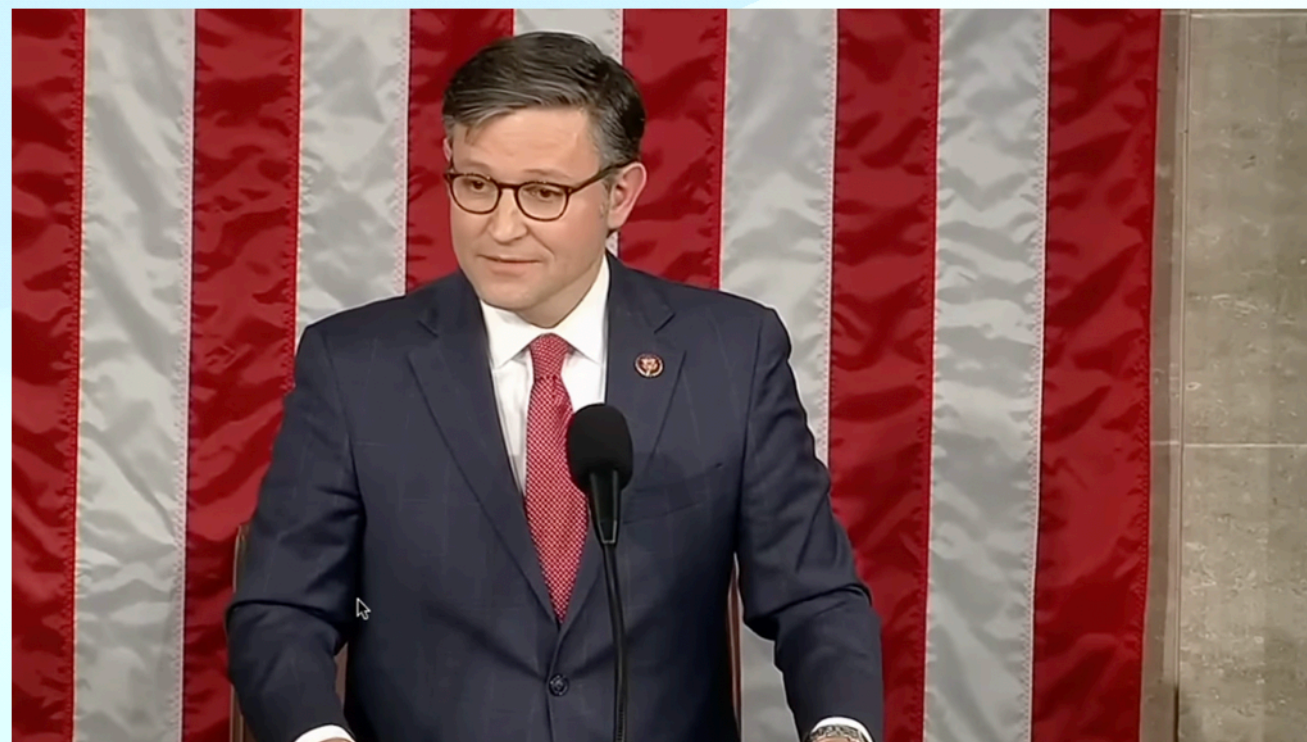
- Floor speech reaches around 850k viewers
- „How“ could be more important than „what“
- Areas of future research
 - Are speakers intentionally „playing with the camera“?
 - Take other measurement for emotional intensity
 - Several other questions remain

Discussion - Personal Opinion

- Can emotional intensity be abused?
 - Probably it can be trained
 - Benefit for „good speakers“
- What is the reason for the third hypothesis only being partially true?
 - Journalists are not involved in the Congress
- What are the goals of congress members?
 - Reach public or Congress Members

Conclusion

Introduction



Questions

- Can emotional intensity be abused by Congress Members?

References

<https://www.semanticscholar.org/paper/This-Floor-Speech-Will-Be-Televised-:-Understanding-Dietrich-Schultz/0b18e9c130373667fcee378df9b452301c3d84d7>

<https://www.youtube.com/watch?v=GCTScJ5twWI&t=76s>

[https://en.wikipedia.org/wiki/United States Congress](https://en.wikipedia.org/wiki/United_States_Congress)